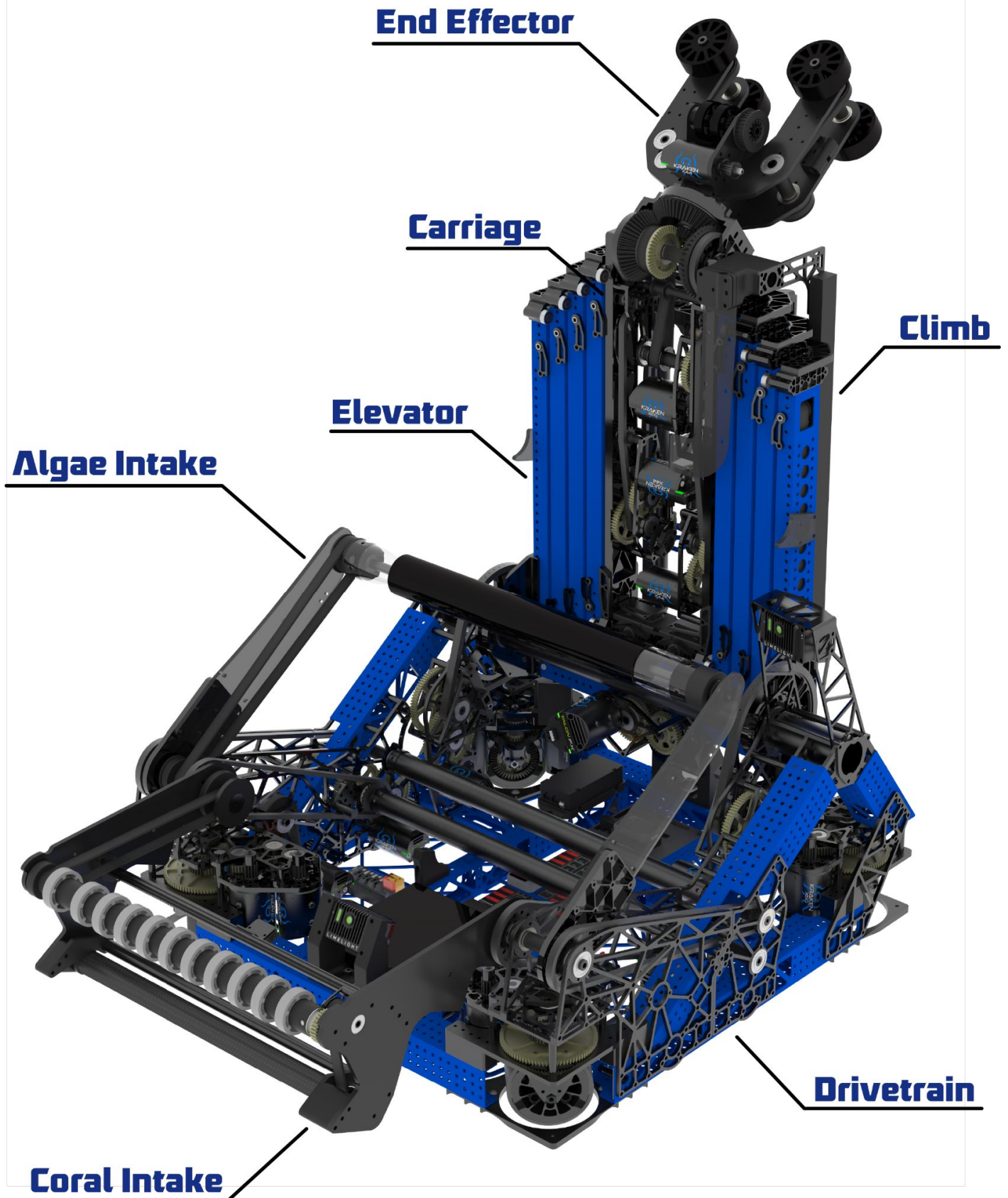




# 2025 Technical Binder

# FINAL DESIGN



# DESIGN FEATURES

## **Drivetrain:**

- Low Center of Gravity
- 28" x 28" perimeter to reduce weight but maintain overall stability

## **Algae Intake:**

- Intake Algae from ground over the bumpers
- Assist in Level 1 scoring

## **Coral Intake:**

- Intake Coral from most angles on ground and funnel to center
- Handoff into End Effector
- Effectively score in Level 1

## **A-Frame & Shoulder:**

- Provide solid support for Elevator when at max extension
- Quickly rotate Elevator from horizontal to vertical in ~1 second

## **Elevator:**

- Four stage continuous elevator with internally ran belts

## **Carriage:**

- Contains gearboxes for differential gearing for End Effector and the End Effector rollers

## **Yolk:**

- Provides solid connection between End Effector and Carriage

## **End Effector:**

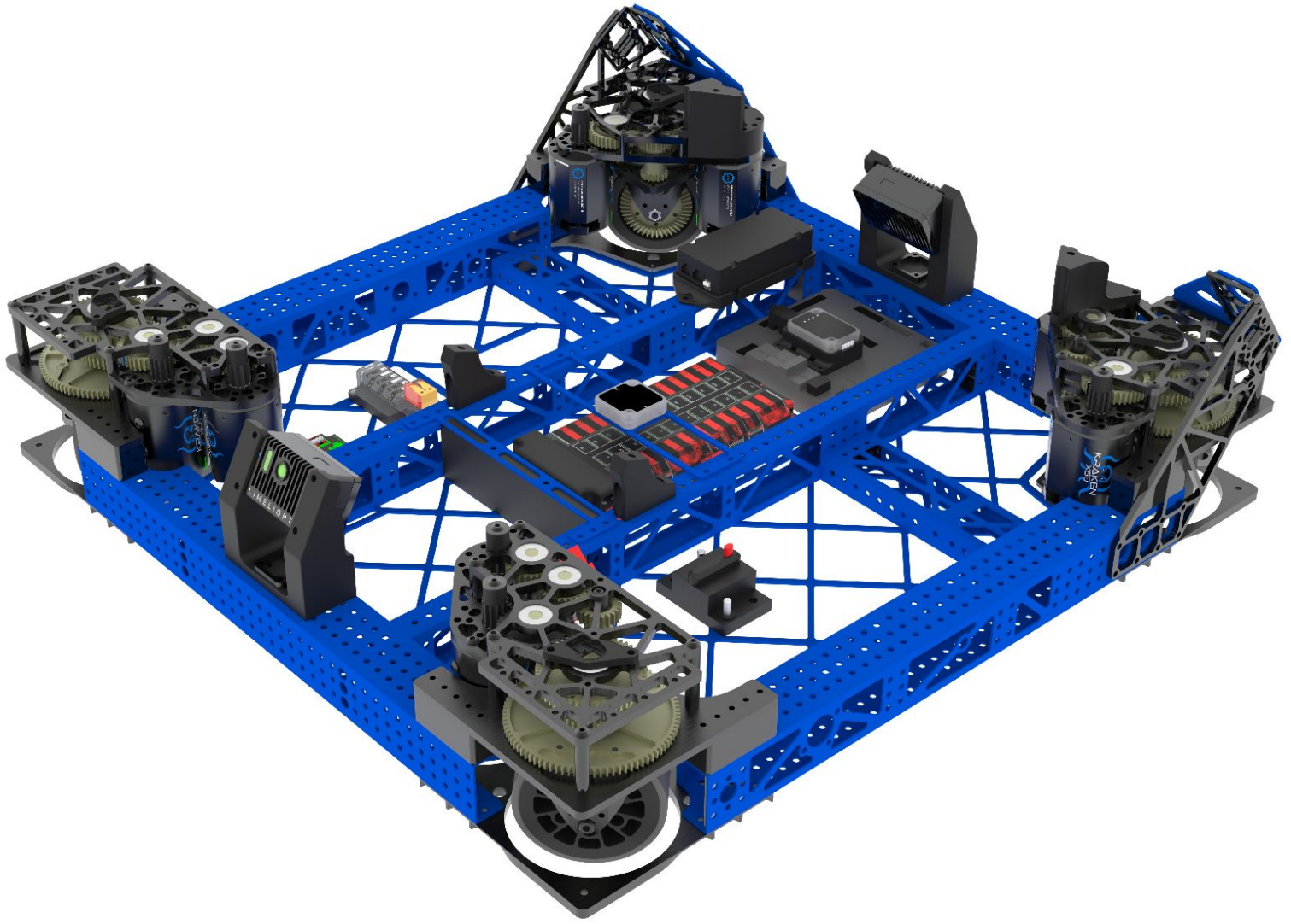
- Intake and score coral on all levels of Reef
- Remove Algae from Reef and place in Barge

## **Climbing System:**

- Uses combination of Elevator, Shoulder, and Algae Intake to perform climb
- Passively slides to reposition Chain to be centered over robot CG



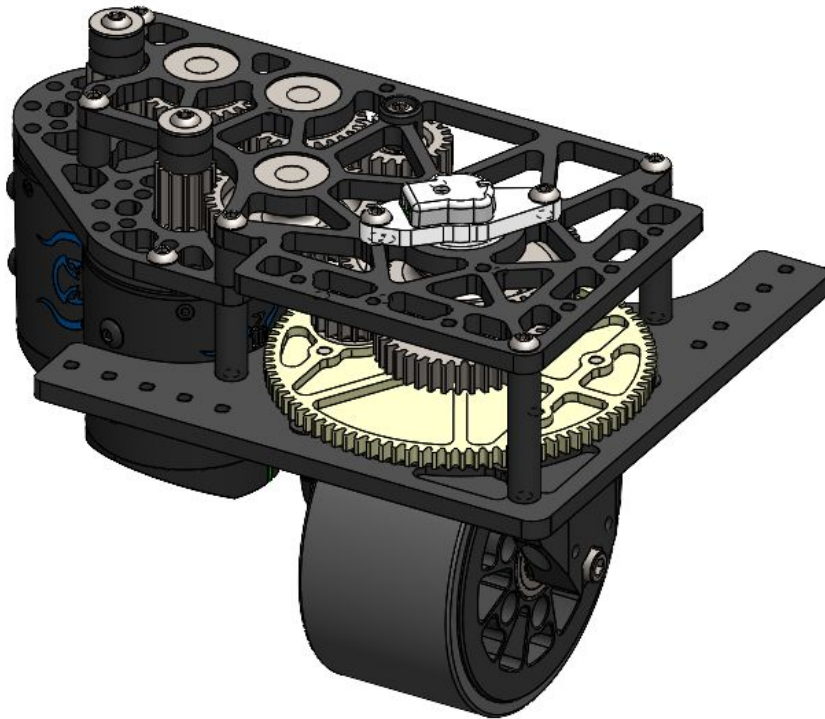
# DRIVETRAIN



## Features:

- 28" x 28" frame size utilizing four 2" x 2" x 1/16" pocketed tubes
- 2" x 1" x 1/16" support tubes span drivetrain length, reinforcing the belly pan and front/rear frame tubes
- .09" belly pan with mounting locations for major electrical components
- WCP Swerve X2 Modules mounted on custom machined 1" tall spacer blocks, lowering drivetrain (~ 0.5" clearance from ground)
  - WCP Swerve X2i used on Elevator side
  - WCP Swerve X2t used on Intaking side

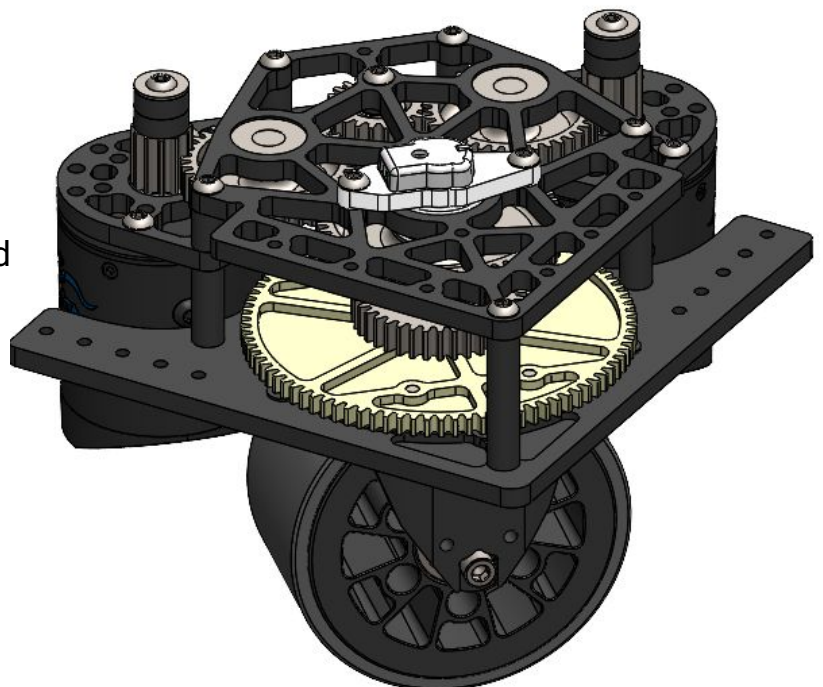
# SWERVE MODULE



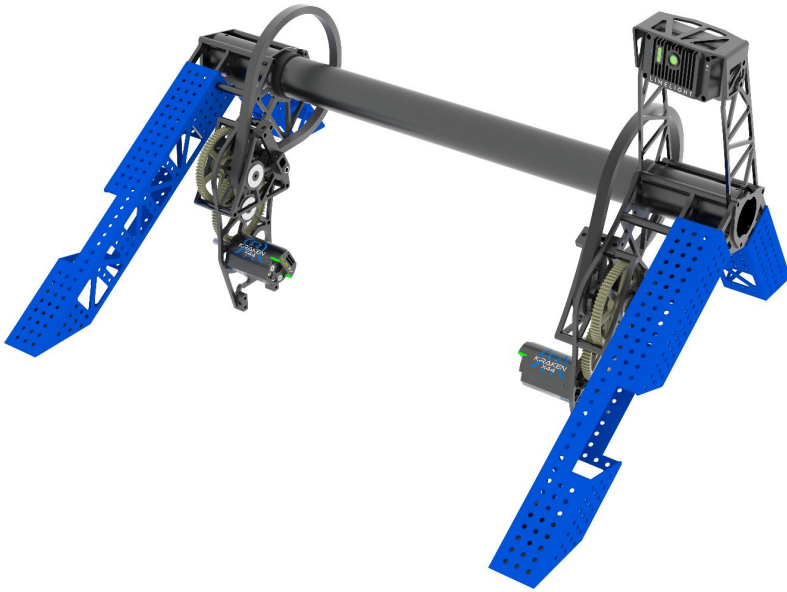
## Features

- X3 gear ratio spread with 10T pinion
- 6.48:1 overall gear ratio (16.1 ft/s free speed)
- Kraken X60 motors used for drive
- Kraken X44 motors used for rotation
  - Custom motor mounting plate to contour X44 better

- WCP Swerve X2i on Elevator side used to protect motors and lower weight than X2t
- WCP Swerve X2t on Intake side used to protect motors and give more clearance for intake gearboxes
- All aluminum gears for weight reduction



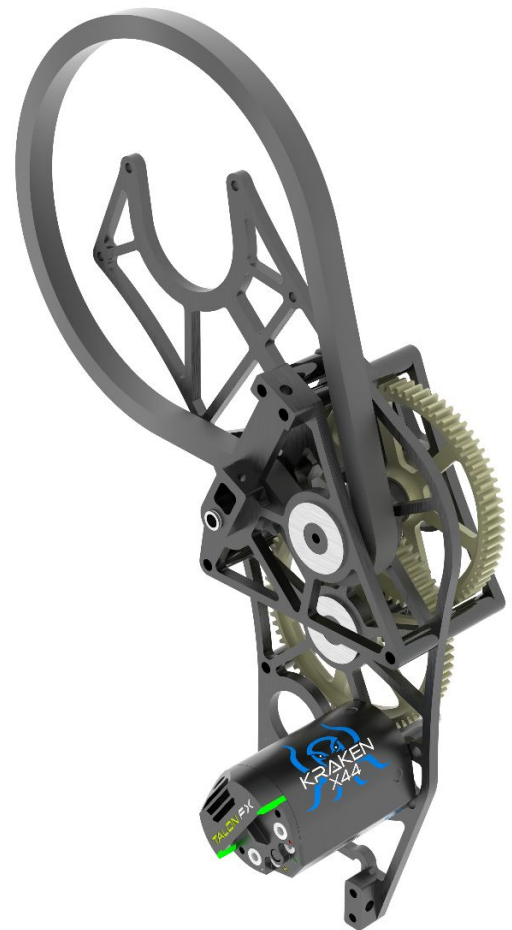
# A-FRAME & SHOULDER GEARBOX



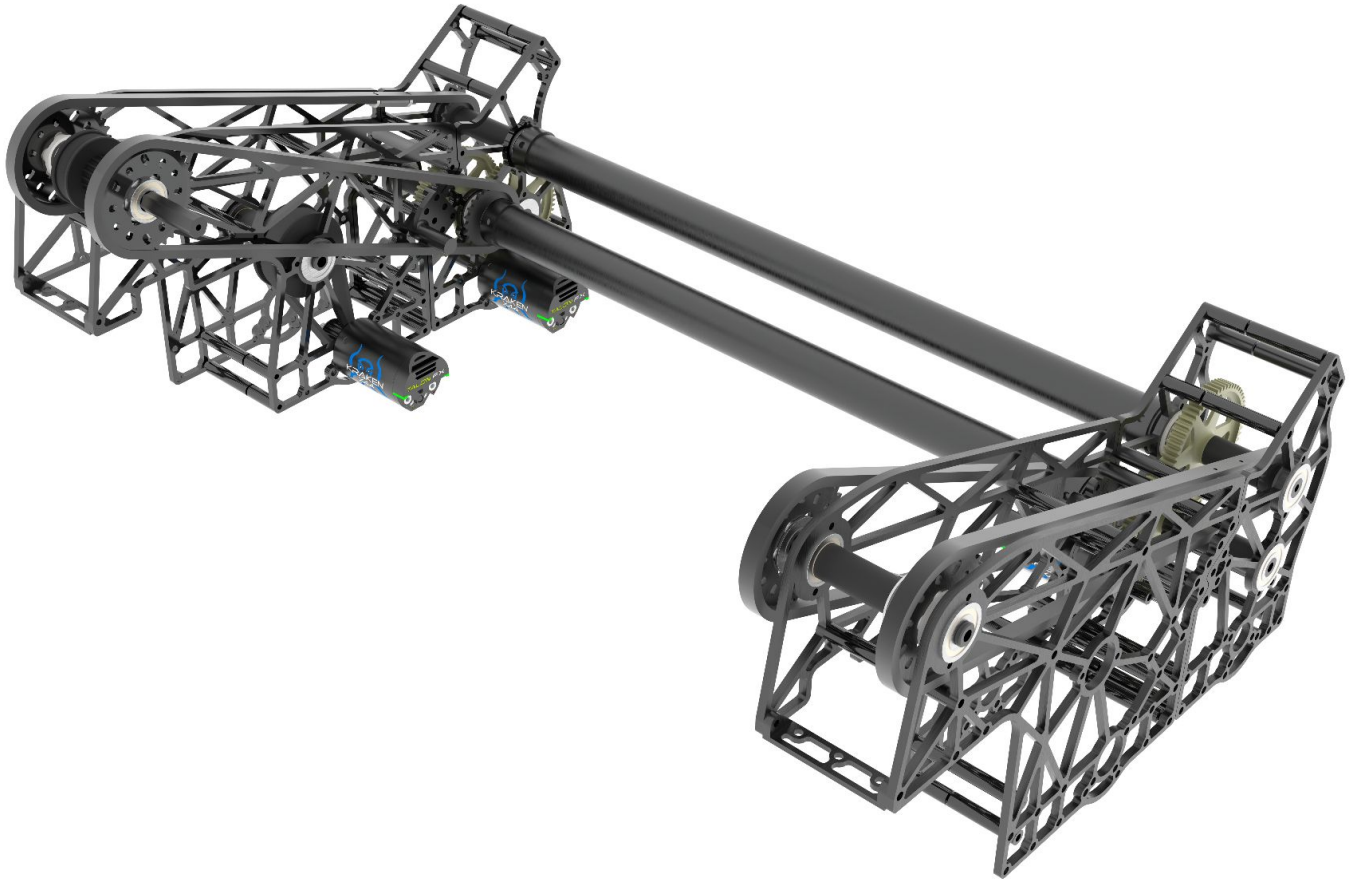
## Features

- Constructed from four 2" x 2" x 1/16" tubes
- Directly mounts to drivetrain and supports intake gearbox
- Limelight 4 positioned for maximum field of view (FOV) in object tracking
  - Shroud to protect from falling Coral
  - Custom Heatsink and fan
- 2" x .03" wall tube used as pivot for elevator

- Custom chain tensioner blocks
- Each gearbox is powered using one Kraken X44 motor
  - Geared at 182.25:1
  - Encoder geared up to be 1:1 with elevator rotation
- Main gearbox plates supported by 2" x 1" tubes found in drivetrain



# ALGAE/CORAL INTAKE GEARBOX



## Features

- Contains all gearing for Algae Intake and Coral Intake
  - Algae Intake pivot geared to 64:1
  - Coral Intake pivot geared to 42.67:1
- Algae Intake and Coral Intake rollers powered together by one Kraken X44 motor
- Algae and Coral Intake pivot about same shaft
- Both intakes are powered on both sides to ensure the intake arms move at the same time
  - Power transferred via two separate transfer tubes



# ALGAE INTAKE



## Features

- Custom billet aluminum arms machined to provide desired rigidity and strength
  - Sprockets keyed into part via MotionX pattern
- WCP SRPP used for Intake plates for maximum durability
- 2" diameter polycarbonate roller with 1/2" round stub axles
  - Cat Tongue grip tape applied to center of roller tube
- Intake is repurposed for supporting drive base when robot is disabled after climbing
  - Latches into End Effector
- Used in combination with Coral Intake for quick and accurate scoring in Level 1



# CORAL INTAKE



## Features

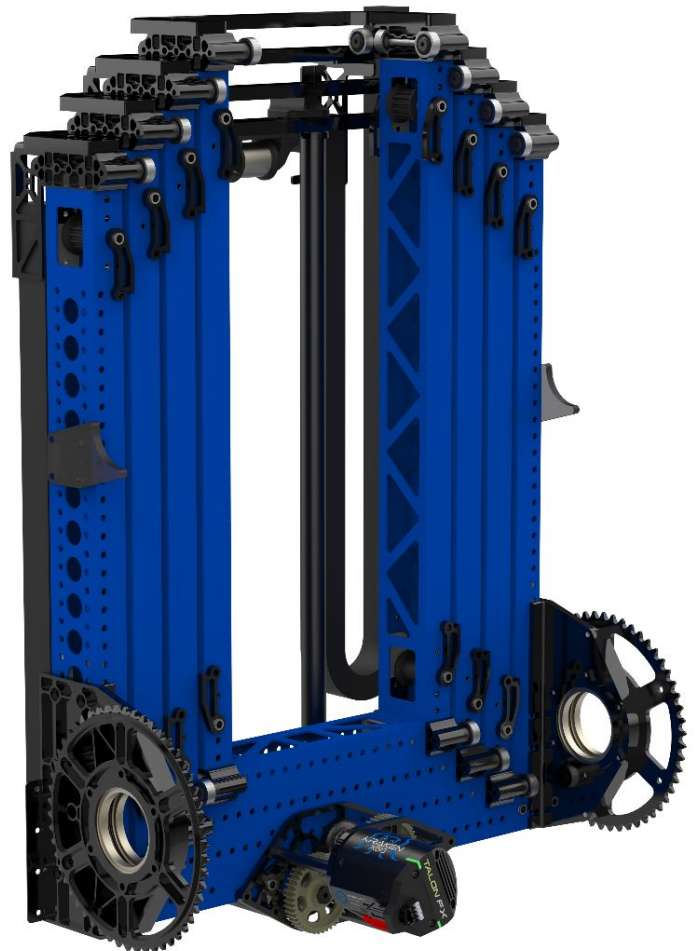
- Polycarbonate roller tubes with WCP Stub Roller Hubs
  - 2" flex wheels for top roller
  - Cat Tongue grip tape for bottom roller
  - ~.125" of compression to hold Coral
- Diffused Proximity Banner sensor for detecting Coral
- WCP SRPP used for Intake plates for maximum durability
- Directly hands off Coral into End Effector
- Passively centers Coral
- Used in combination with Algae Intake for quick and accurate scoring in Level 1

# ELEVATOR

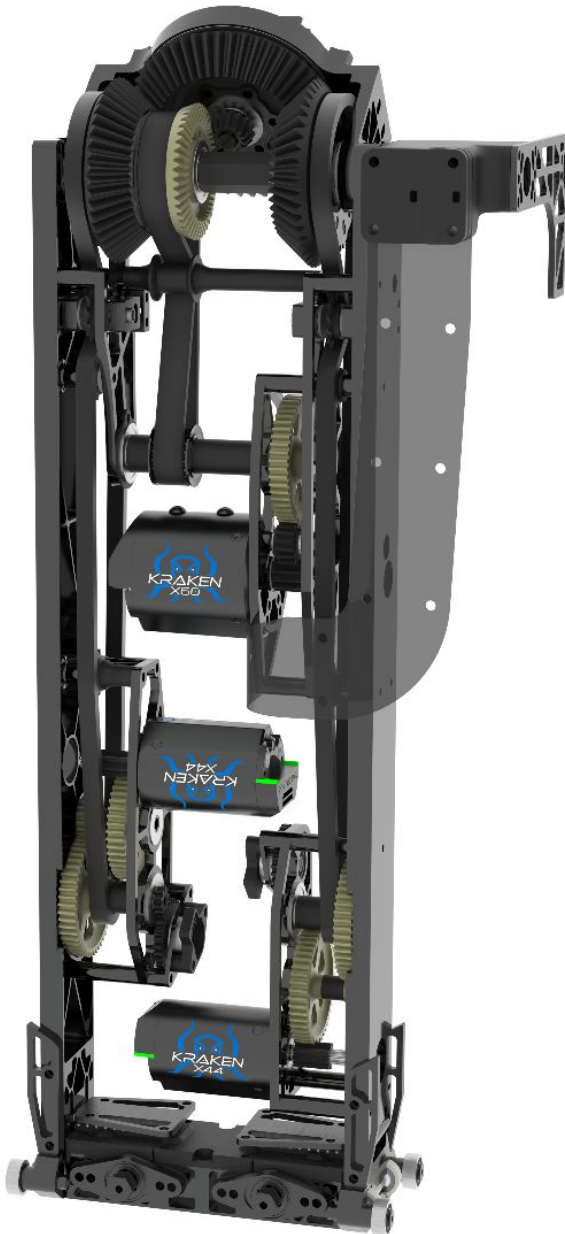


## Features

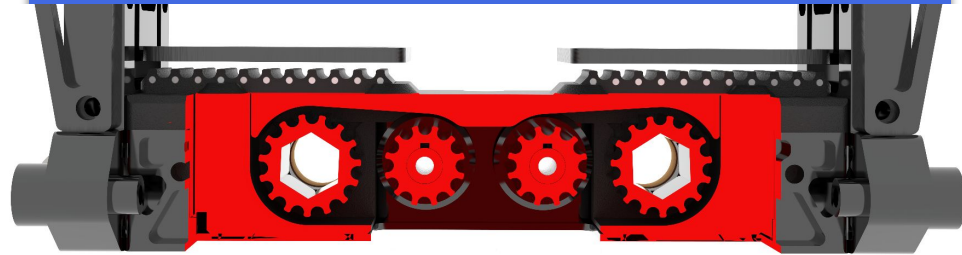
- Four stage continuous elevator
  - Two internally ran 18mm wide 5mm pitch HTD belts
- Billet cross bars for increased rigidity
- Powered by one Kraken X60 motor
  - Two stage gearbox geared at 6.875:1
- Interlocking hard stops on each stage increase rigidity when Elevator is at max extension
- Can reach net for precise placement of Algae



# CARRIAGE



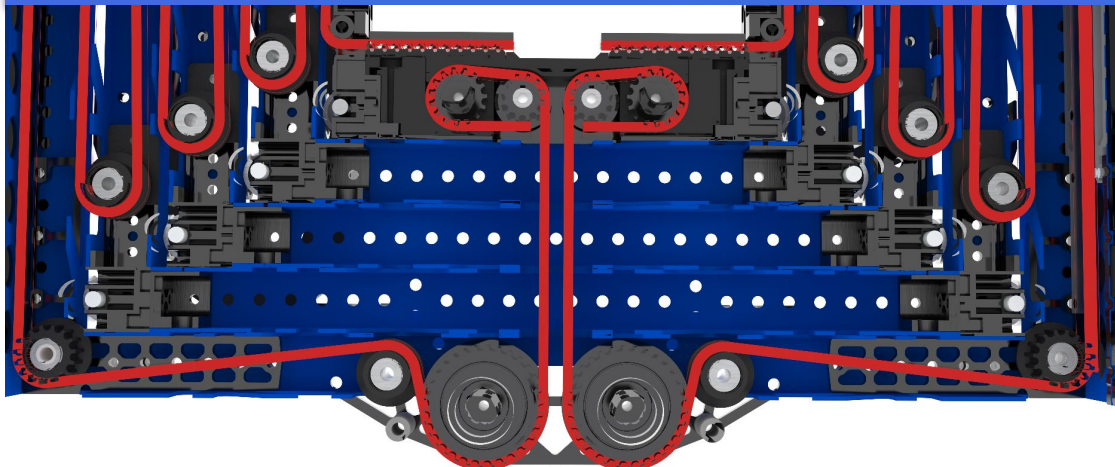
## BELT TENSIONER



### Features

- Billet carriage plates with three gearboxes integrated into them for reduced part count
- Custom low profile belt tensioner integrated at the bottom of carriage for return belt
  - Belt is ran through tensioner via two pulleys and is held down by a 3d printed cover
  - WCP ratchet plate maintains tension of belt

## SECTION VIEW OF RIGGING & TENSIONER



# DIFFERENTIAL GEARBOX & YOKE



## Features

- Custom 3D-printed bevel gears driving a differential gearbox to rotate End Effector
  - Bevel gears independently powered through identical three-stage 61.6:1 gearbox
- Custom billet Yoke bearing housing for output bevel gear contains two 3.5" X-contact bearings

- Powered via gearboxes located within Carriage
  - Final reduction done via #25H chain
- Machined in hard stops to prevent over rotation of differential gears





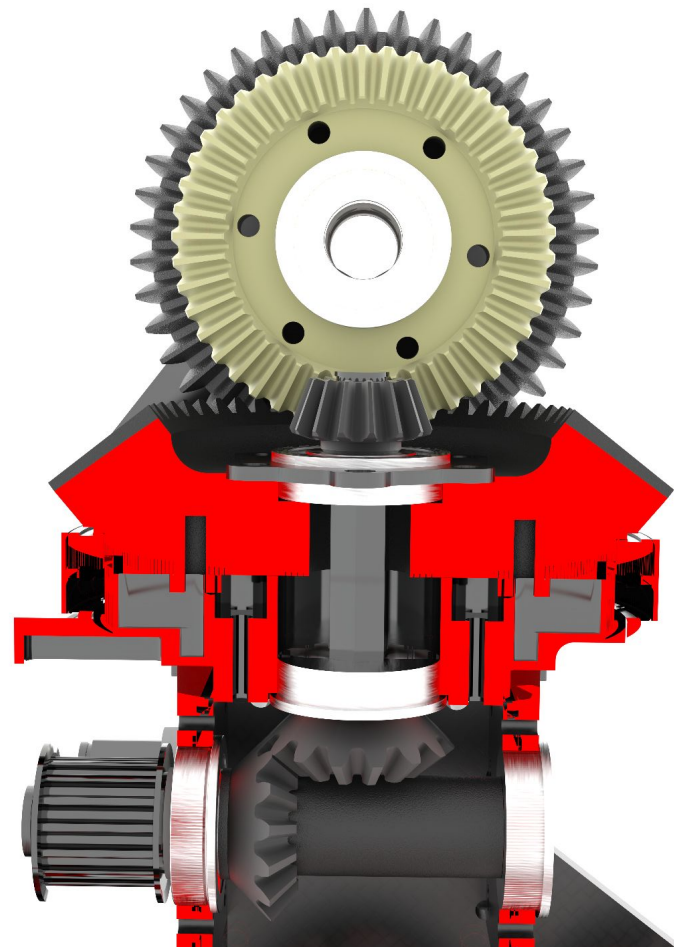
# END EFFECTOR



*Section View*

## Features

- Two sets of independently driven flex wheels enable both horizontal and vertical manipulation of coral
- Front 3" flex wheels capable of pinching algae for scoring
- Front flex wheels powered by gearbox located in Carriage
  - Powered through center of differential gears
  - Allows the End Effector to stay as light and compact as possible
- Vertical rollers are powered by a Kraken x44 motor at a 3:1 ratio



# CLIMBING SYSTEM



## Features

- Two 3" 45A flex wheels to bring vertical pipe of Cage into latching mechanism
  - Powered by one Kraken X44 motor
- Spring loaded latches to prevent Cage from slipping out of mechanism
- Diffused Proximity Banner sensor to sense when Cage is latched
  - Initiates automated climbing sequence
- Linear bushings for smooth sliding on  $\frac{3}{4}$ " Aluminum tubes
  - Sprung up with constant force spring
  - Pulled down by motion of Elevator folding during climbing sequence